Quantification: 542_z18
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3688.27	3684	1441.75	px^2
major_axis_length_px	lacuna shape	132.275	128.431	35.9804	px
minor_axis_length_px	lacuna shape	35.8241	35.4482	8.8535	px
aspect_ratio	lacuna shape	3.6839	3.6773	0.6253	unitless
eccentricity	lacuna shape	0.9593	0.9623	0.0137	unitless
solidity	lacuna shape	0.8643	0.893	0.0736	unitless
roots_count	headline	16.1818	14	6.7944	count
ring_length_r30_px	headline	517.909	557	174.739	px
ring_length_r60_px	reported	1210.09	1284	380.911	px
owned_length_px	ownership-dependent	3112.86	3118.08	1009.04	px
edge_count	ownership-dependent	143.182	157	51.8726	count
mean_edge_length_px	ownership-dependent	22.6114	21.6578	3.7876	px
perimeter_px	normalised	349.688	337.706	124.859	px
ring_area_r30_px2	normalised	11901.2	12118	3231.07	px^2
ring_area_r60_px2	normalised	28359.4	29514	6840.82	px^2
in_frame_fraction_r30	normalised	0.9834	1	0.0361	unitless
in_frame_fraction_r60	normalised	0.9555	1	0.0828	unitless
ring_density_r30	normalised	0.0427705	0.042351	0.00729128	px^-1
ring_density_r60	normalised	0.0419393	0.0435398	0.0073817	px^-1
roots_per_100px_perimeter	normalised	4.7828	4.5367	1.5172	per 100 px
ring_attached_length_r30_px	network v2	481.546	505	183.009	px
ring_attached_length_r60_px	network v2	1050.45	1115	402.904	px
ring_length_w_r30_px	network v2	583.357	624.831	201.165	px
ring_length_w_r60_px	network v2	1369.62	1442.92	436.311	px
sholl_crossings_r10	network v2	13	14	5.1962	count
sholl_crossings_r20	network v2	16.0909	17	6.1066	count
sholl_crossings_r30	network v2	16.8182	17	5.3445	count
loop_count	loops, report only	12.6364	14	6.3761	count
loop_edge_share	loops, report only	0.3503	0.3235	0.1114	share
ring_width_mean_r30_px	width	7.5362	7.4935	0.5369	px

Per field
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measure	value	units
analysed_area_px2	1.00717e+06	px^2
total_skeleton_length_px	39855	px
canalicular_length_density_per_px	0.0395713	px^-1
junction_count	923	count
junction_density_per_px2	0.00091643	px^-2
lacuna_count	13	count

lacunae_per_px2	1.291e-05	px^-2
skeleton_component_count	228	count
mean_component_length_px	174.803	px
median_component_length_px	22.5	px
field_length_density_w_per_px	0.0447528	px^-1
field_density_without_flagged_per_px	0.0395294	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	372	px
unattached_component_count	22	count
canalicular_length_density_attached_per_px	0.0395713	px^-1
field_length_density_with_unattached_per_px	0.0399407	px^-1
loop_count	170	count
loop_edge_share	0.345	share
edge_count	1997	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2487	px
width_median_px	8.2462	px
width_p10_px	4.4721	px
width_p90_px	10.7703	px
width_pixels_used	38265	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	602	56.2113	21.2037	2.651	0.9261	0.6645	-0.3574	16.2608	914.042	True
2	5924	181.043	48.2611	3.7513	0.9638	0.675	-0.0924	129.871	638.627	False
3	4438	160.681	35.4482	4.5328	0.9754	0.8989	-0.2181	173.351	382.514	False
4	4342	127.796	43.8553	2.914	0.9393	0.893	-0.1983	210.3	874.108	False
5	5254	148.687	46.2758	3.2131	0.9503	0.9227	-0.0688	369.252	79.7008	False
6	3405	124.602	35.4524	3.5146	0.9587	0.8958	-0.0028	532.24	401.315	False
7	4133	149.961	37.557	3.9929	0.9681	0.861	-0.1428	589.071	924.049	False
8	2572	119.294	30.2433	3.9445	0.9673	0.8307	-0.0493	570.032	785.133	False
9	3684	159.337	32.9692	4.8329	0.9784	0.8054	-0.098	739.452	247.536	False
10	3421	128.431	34.9252	3.6773	0.9623	0.8774	-0.2145	813.822	732.524	False
11	2890	110.803	33.408	3.3167	0.9535	0.9299	0.0034	944.816	407.904	False
12	508	44.3947	15.6697	2.8331	0.9356	0.917	-0.2083	990.155	937.699	False
13								1015.14	933.902	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	241	606	669.144	38	17.6091	6.9616
2	28	754	1600	3114.14	157	19.8353	7.4838
3	14	557	1287	3725.14	172	21.6578	8.0727
4	26	626	1412	4682.35	227	20.6271	7.1625
5	12	500	1178	2475.51	131	18.897	8.0756
6	14	479	1252	3953.39	170	23.2552	8.0328

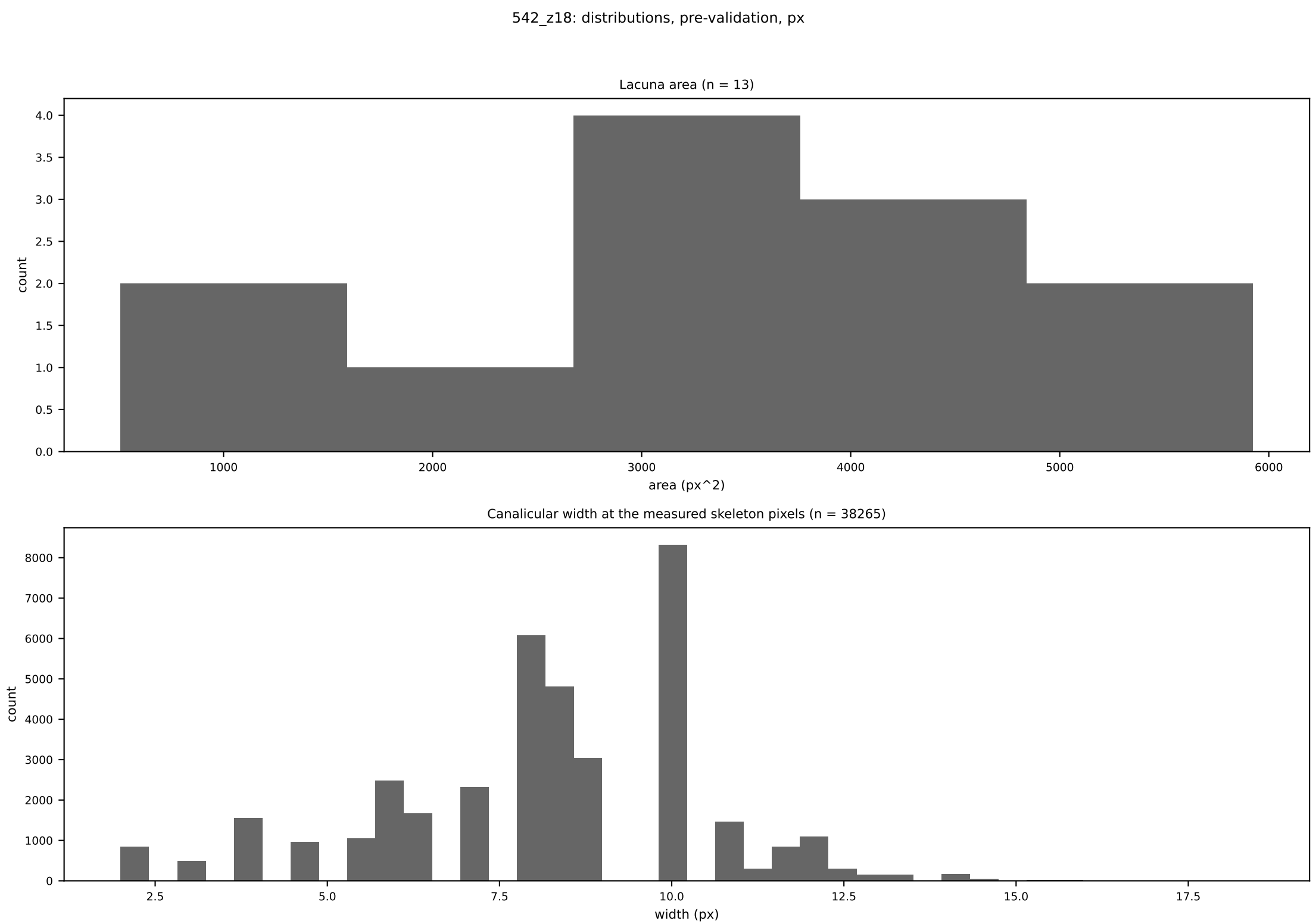
7	20	683	1638	3118.08	161	19.367	8.0953
8	14	497	1139	3528.26	142	24.8469	7.4935
9	22	608	1284	2913.58	151	19.2953	7.0786
10	11	566	1463	3797.36	163	23.2967	7.0826
11	9	288	708	1722.92	55	31.3259	7.832
12	8	139	350	1210.77	46	26.3212	6.4883
13	0	1	44	0	0	0	6

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	142.811	4575	11641	0.673	0.6207	0.0526776	0.0520574	2.8009
2	625.725	16860	35430	0.9542	0.8861	0.0447212	0.0451595	4.4748
3	377.019	13152	31926	1	1	0.042351	0.040312	3.7133
4	337.706	12118	29514	1	1	0.0516587	0.0478417	7.699
5	371.806	13199	31872	1	0.9956	0.0378817	0.0369603	3.2275
6	308.593	11379	28385	1	1	0.0420951	0.0441078	4.5367
7	382.877	12760	30245	1	1	0.0535267	0.0541577	5.2236
8	306.492	10697	26160	1	1	0.0464616	0.0435398	4.5678
9	437.262	14880	35351	1	1	0.0408602	0.0363215	5.0313
10	326.836	11837	29271	1	1	0.0478162	0.0499812	3.3656
11	263.48	9968	22850	0.9798	0.8783	0.0288925	0.0309847	3.4158
12	108.77	4063	10949	0.8837	0.75	0.0342112	0.0319664	7.355
13	59.8995	971	1939	0.6158	0.5712	0.00102987	0.0226921	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	234	478	274.551	682.872	5	7	6
2	734	1304	857.11	1827.96	20	24	23
3	505	1109	624.831	1442.92	13	17	19
4	608	1214	705.357	1619.55	16	20	20
5	403	981	559.304	1333.19	7	14	17
6	449	1104	540.375	1433.04	11	15	17
7	670	1572	767.5	1836.3	19	24	20
8	497	1115	553.919	1280.21	14	16	16
9	578	1117	699.156	1464.31	16	17	17
10	518	1419	640.701	1652.25	16	18	22
11	211	374	322.664	793.097	7	9	8
12	124	246	146.012	383.007	4	3	6
13	0	0	1.4142	48.2132	0	0	1


542_z18: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 38265)
count
width (px)